

Implementation of Nitrogen Removal at Detroit WRRF

ERIE ENVIRONMENTAL
ENGINEERING

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Meet the Team

Riley McGaughey



Project Engineer

Dylan Rian Gong



Design Engineer

Morgan Smith



Environmental Consultant

Problem Scope

- Future DWRRF limits more stringent
- Effluent total nitrogen (TN) $\leq$ 3 mg/L
 - 659 MGD avg daily flow
- Large-scale advanced N removal less common
- Cost versus performance

Current NPDES limit parameters	Detroit River Outfall	Rouge River Outfall
CBOD ₅ (mg/L)	40	25
TSS (mg/L)	70	30
TP (mg/L)	1.5	0.6-0.7
Ammonia-N (mg/L)	Report	Report
pH	6.0-9.0	6.0-9.0
Total residual chlorine (ug/L)(1)	0.11	0.038
Min dissolved oxygen (DO) (mg/L)	Report	3.0

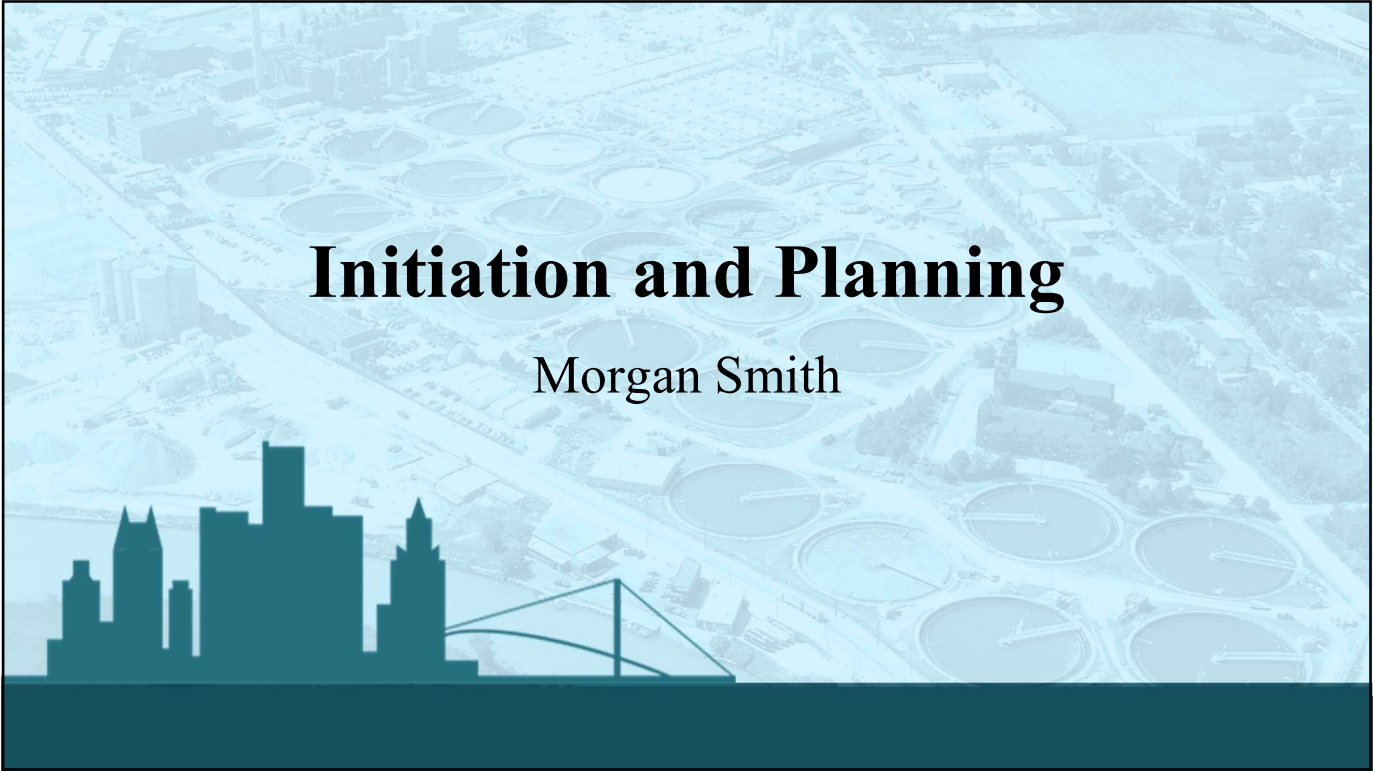
*(1)=Daily limit

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Components of Project

- Initiation and planning
 - Existing DWRRF system and key considerations
- Development of N removal system
 - SUMO24 proposed treatment train
 - Analysis of system performance
- Evaluation of impacts and outcomes
 - Implementation considerations
 - Conclusions and next steps

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Initiation and Planning

Morgan Smith

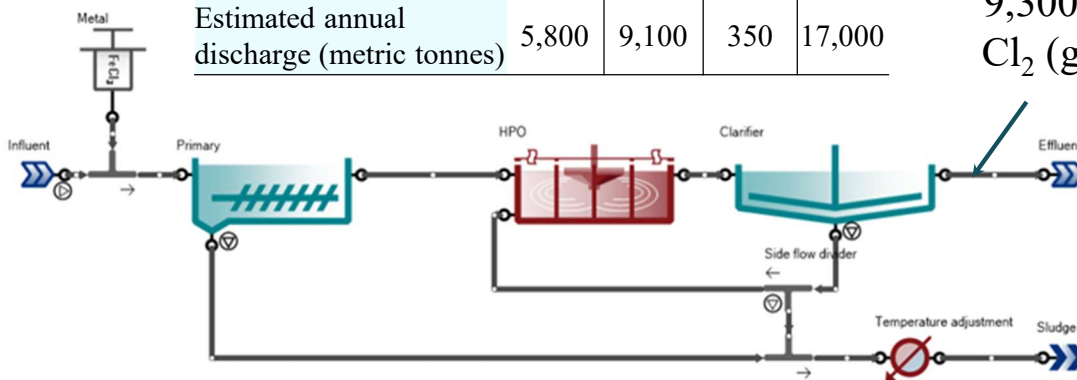
Project Goals

- Compliance with limits and regulations
- Evaluate innovative treatment methods
- Anaerobic digester (AD) energy generation
- Optimization:
 - Tank hydraulic retention time (HRT): 6-12hrs
 - Nitrification sludge retention time (SRT): 5-20d
 - Digester SRT: 10-70d
 - Carbon requirements

Current DWRRF System and Baseline Data

Provided parameters	BOD ₅	TSS	TP	TN
Avg secondary effluent concentration (mg/L)	5.8	9.0	0.39	17
Estimated annual discharge (metric tonnes)	5,800	9,100	350	17,000

9,300 lb/day of Cl₂ (gas) added



Key Design Considerations

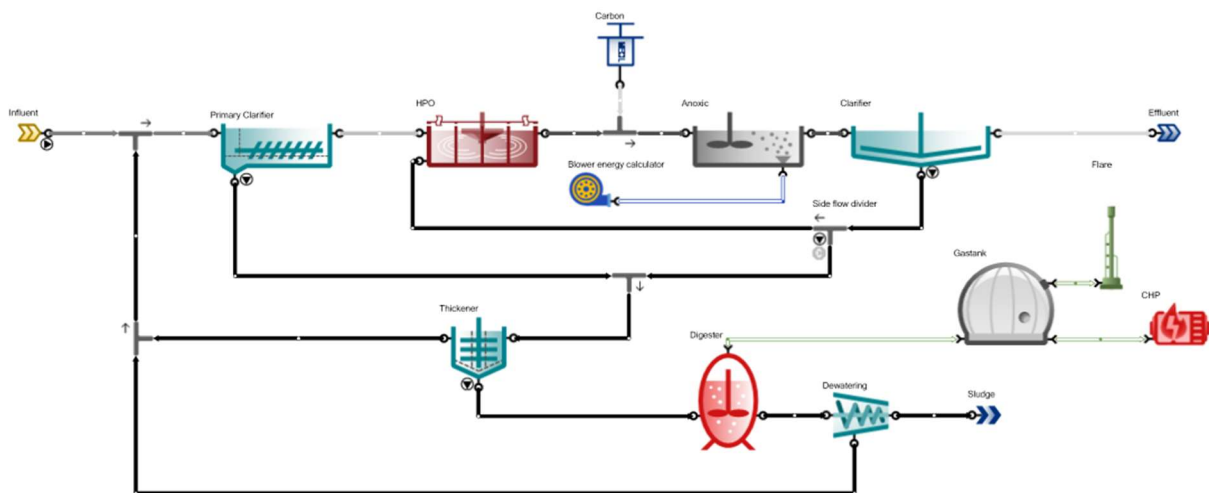
- Integration into plant
- Cold climate performance
- Microbial growth conditions
- Minimized chemical additions

Parameter	Avg daily influent value	Unit
Total BOD ₅	86	mg O ₂ /L
Particulate BOD ₅	41.7	mg O ₂ /L
Total Kjeldahl nitrogen (TKN)	16.5	mg N/L
TP	2.3	mg P/L
Temperature	15.83	°C

Development of N Removal System

Dylan Rian Gong

Proposed DWRRF Treatment Train



Selection Rationale

- High purity oxygen (HPO) system preserved
 - Minimal infrastructure changes
- Conventional nitrification-denitrification
 - Return sludge utilized

Obtained parameters	BOD ₅	TSS	TP	TN
Proposed avg secondary effluent concentration (mg/L)	2.5	10.0	1.07	2.63
Estimated annual discharge (metric tonnes)	2,500	10,100	1,030	2,630

- 84.5% N removal efficiency achieved
 - Reliable performance



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Proposed System Characteristics

→ Biogas generation
~48,339 kWh/d

Parameter	Value			Unit
Effluent CO ₂	81			mg TIC/L
Effluent DO	0.00043			mg O ₂ /L
	Nitrification-Denitrification	Anaerobic digester		
SRT	13	—		days
HRT	0.259	13.4		days
	HPO Aerobic clarifiers	Anoxic clarifiers		
Number of units	4	1	1	—
Depth	10	5	10	m
Volume	100,000	250,000	10,000	m ³
DO setpoint	6, 4, 2, 0	0	—	mg O ₂ /L

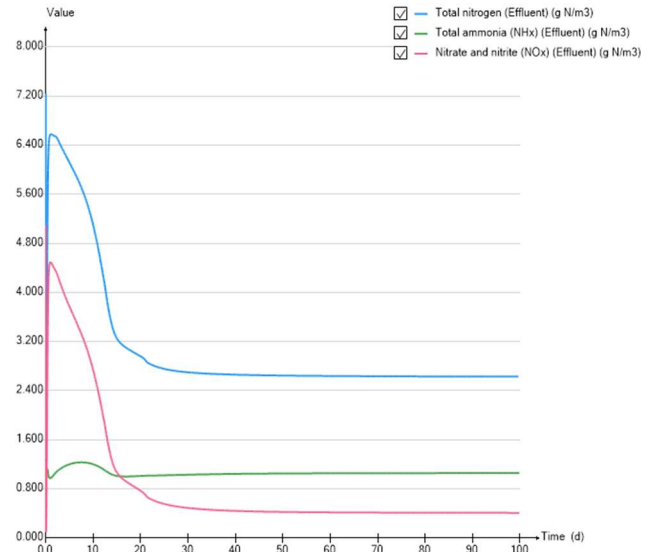
Parameter	Sludge processing	Unit
Drying facilities	750	m ³ /d
Recycled	23,750	m ³ /d
Gas production		
Biogas production	24,169.28	m ³ /d
Electricity		
Sludge stream only	550.1	kWh/d
Entire system	380,583.04	kWh/d
Chemical consumption and recovery		
CH ₃ OH consumption	31,680	kg CH ₃ OH/day
Struvite recovery	115,739	kg/day



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Performance Assessment

- TKN: 16 to 2.2 mg N/L
- NH_x : 12 to 1.1 mg N/L
- Steady state after 25 d
 - 100d interval for SUMO24 evaluation



Codes and Regulations

- NPDES permit #MI0022802
 - Effluent → Detroit and Rouge Rivers
 - Limits and reporting requirements
- CWA Section 303(d)
 - Total maximum daily loads (TMDLs)
- 40 CFR Part 503
 - Land applications standards and incineration standards
 - Regulation of Biosolids Drying Facility
- CAA Part 55 of MI Act 451
 - Title V air permit for emissions

Preliminary Cost Estimates

- Industrial avg electricity price (USD)= 8.6¢/kWh
 - \$32,731/d
 - Biogas production= \$4,158/d
- Methanol price (USD)= \$~600/MT
 - \$19,008/d

Capital cost (US \$/gal/d)	O&M cost (US \$/lb)	Life-cycle costs (US \$/lb)
\$3.32	\$0.13	\$3.45

- Estimates of \$10M-\$25M for digester construction
- 6-16 year payback period

Evaluation of Impacts and Outcomes

Riley McGaughey

Technical Feasibility

- Economic strain on nearby communities
 - Environmental benefits
 - Decreased nutrient loading
 - \$ of prevention < \$ of remediation
 - Shift to circular systems
 - Minimized construction space needed
 - Vacant lots available
- Ethical considerations take precedent
- Value of human and ecosystem health



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External Funding

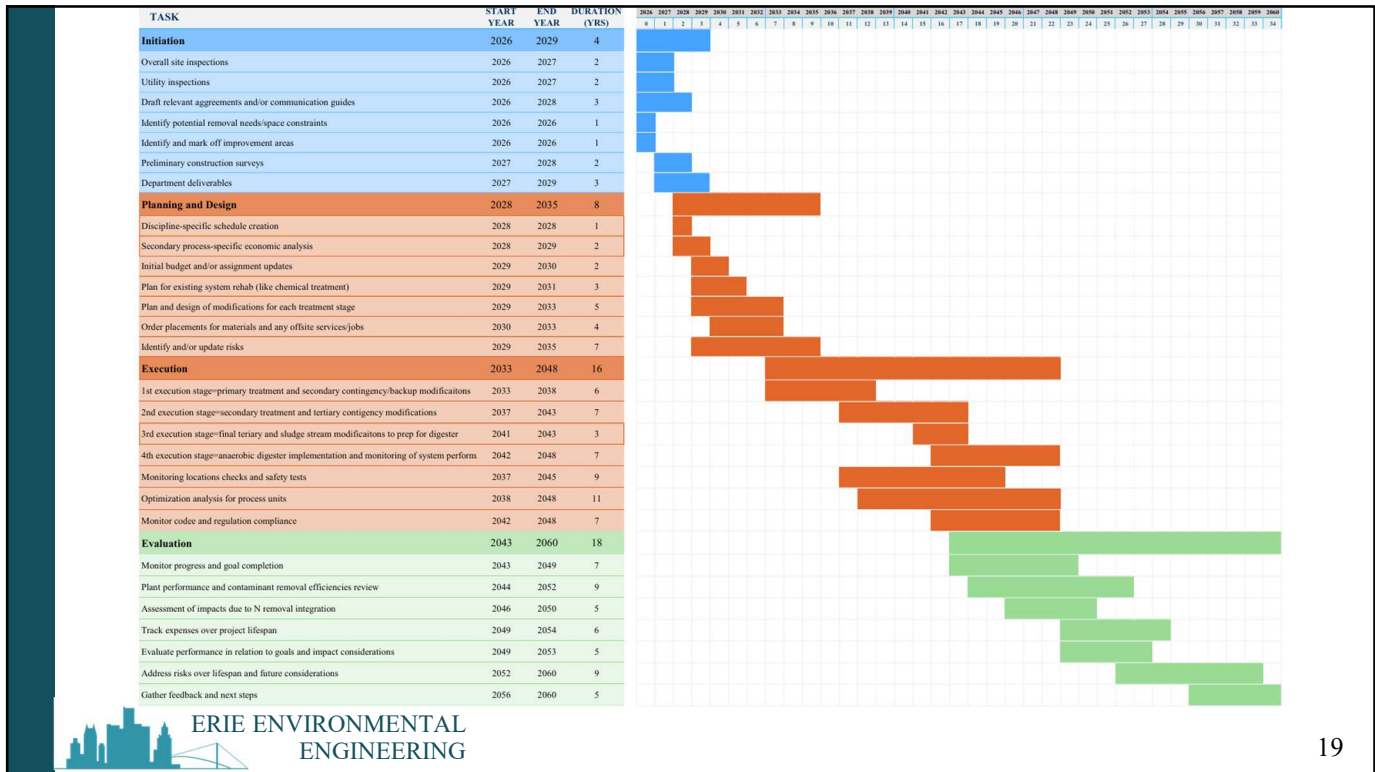
- Support for implementation
- Water Infrastructure Finance and Innovation Act (WIFIA)
- Other federal grants

\$10M to fund startup



Database: Grants.gov		Project #: 495	
Grant Programs		<i>All fields are required and must be completed to be eligible for funding</i>	
Proposal Coversheet			
Organizational Information:			
Organization:	Erie Environmental Engineering		
Address:	1534 W Jefferson Ave, Unit 9		
City:	Detroit	State:	MI Zip: 48209
Organizational Leader:	Riley McGaughey	Title:	Project Manager
Email:	mcgaug17@msu.edu		
Telephone:	(810) 542-1445	Fax:	n/a DUNS #: n/a
Project Contact Information (if different from above)			
Project Contact Name:	Dylan Rian Gong	Telephone:	(517) 525-2752
Project Contact Title:	Design Manager	Email:	gongdyla@msu.edu
Project Contact Address:	1534 W Jefferson Ave, Unit 9		
City:	Detroit	State:	MI Zip: 48209
Project Information:			
NSF Funds Required:	\$10,000,000.00	Grant Program:	U.S. National Science Foundation
Project Title:	Nitrogen Removal Additions to DWRWF Treatment Train		
Brief description of project purpose and goals (Limit 250 characters, including spaces)			
Incorporation of nitrogen removal processes into the existing DWRWF treatment train and modifications with the goal of achieving a year-round final effluent TN concentration of ≤ 3 mg/L. This will greatly decrease Rouge River eutrophication concerns.			
Funding Opportunity #:	PD-20-1440	Start Date:	1/14/2026 End Date: 1/31/2030

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Conclusions

- Nitrification-denitrification
- Effluent N goal achieved
 - TP, BOD₅, TSS also decreased
- Revenue from AD
 - Less dependence on varied retail electricity \$
- Up front economic impacts
- Enhanced water quality in receiving waters
- Paves way for planned GLWA improvements

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Next Steps

- Pilot program 
- Sludge management improvements
 - Handling and end-use
- Further investigation of funding options
 - Collaborative research
- Implementation of system
- Thermal hydrolysis processing system (THP) with AD
- Optimized chemical additions

Conclusion

Michigan State University
College of Engineering
CE 495-Senior Capstone Design
Erie Environmental Engineering

**An extra special thanks to the judges,
faculty, and audience members, for
being with us here today!**

An aerial photograph of a wastewater treatment plant, featuring numerous large circular aeration tanks arranged in a grid. In the foreground, a dark silhouette of a city skyline is visible against a teal background. The word "Questions?" is centered in a large, bold, black serif font.

Questions?

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